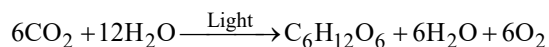


Photosynthesis in Higher Plants

SHORT NOTES

- ❖ Green plants prepare their own food by photosynthesis.
- ❖ Photosynthesis is an anabolic, endergonic and oxidation-reduction process.



- ❖ Photosynthesis takes place mainly in leaves- mesophyll cells have large number of chloroplasts.
- ❖ It occurs in two stages- light reaction and dark reaction.
- ❖ **Light reaction**
- ❖ Within chloroplast, the membrane system is responsible is responsible for trapping light energy and synthesis of ATP and NADPH takes place.
- ❖ Light energy is absorbed by the pigment systems (PS I and PS II).
- ❖ PS I has an absorption peak at 700 nm – P700 and PS II has an absorption peak at 680 nm- P680.
- ❖ After absorbing light, electrons become excited and transferred through PSII to the Electron transport system and then to PS I and finally to NADP^+ forming NADPH.
- ❖ Splitting of water is associated with PS II and results in the release of half molecule of oxygen, two protons and two electrons.

- ❖ During transfer of electrons through ETS, a proton gradient is created across the membrane of thylakoid.
- ❖ Breakdown of this gradient by movement of protons through ATP synthase results in the synthesis of ATP.
- ❖ **Dark Reaction (carbon reactions)**
- ❖ During this process, ATP and NADPH (light reaction products) are utilized for assimilation of CO_2 to Carbohydrates.
- ❖ This reactions takes place in stroma of chloroplast.
- ❖ This is the biosynthetic phase of photosynthesis.
- ❖ It occurs through Calvin cycle, C_4 pathway and CAM.
- ❖ RuBisCO also catalyses a wasteful processt in C_3 plants called Photorespiration.
- ❖ Photorespiration is absent in C_4 plants.
- ❖ Crassulacean Acid Metabolism (CAM) Pathway is a special type of carbon fixation pathway it is observed in plants that grow in dry, semi-arid, or xerophytic conditions,
- ❖ In order to survive in dry conditions, plants adapt to the CAM pathway.
- ❖ Both external (Light, temperature, CO_2 concentration, water) and internal factors (number, size, age and orientation of leaves, mesophyll cells and chloroplast) affects the rate of photosynthesis.